Medical Expense Prediction

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Purpose:

- A healthy body and mind become healthy life.
- Cost of treatment is what we will try to explore and decide for various patients.
- Many reasons are there for medical expenses such as age, city of residence, type of clinic, and diagnosis.

Libraries:

```
import numpy as np
import pandas as pd
import os
import matplotlib.pyplot as pl
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
data = pd.read_csv('../input/insurance.csv')
```

```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import PolynomialFeatures
from sklearn.metrics import r2_score,mean_squared_error
from sklearn.ensemble import RandomForestRegressor
```

First look on the dataset:

- Luckly, no missing content was found which increases the veracity of this dataset.

```
data.head()
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

```
data.isnull().sum()
```

```
age         0
sex         0
bmi         0
children    0
smoker      0
region      0
charges     0
dtype: int64
```

Label Encoding:

- Since we need to predict the amount of costs, we will encode the categorical features.

```python
from sklearn.preprocessing import LabelEncoder
#sex
le = LabelEncoder()
le.fit(data.sex.drop_duplicates())
data.sex = le.transform(data.sex)
# smoker or not
le.fit(data.smoker.drop_duplicates())
data.smoker = le.transform(data.smoker)
#region
le.fit(data.region.drop_duplicates())
data.region = le.transform(data.region)
```

- In general, categorical variables with large variability are best encoded using OneHotEncoder.
- However, in this case, since there is no special order in which the regions would be listed, nothing will change.

Correlation:

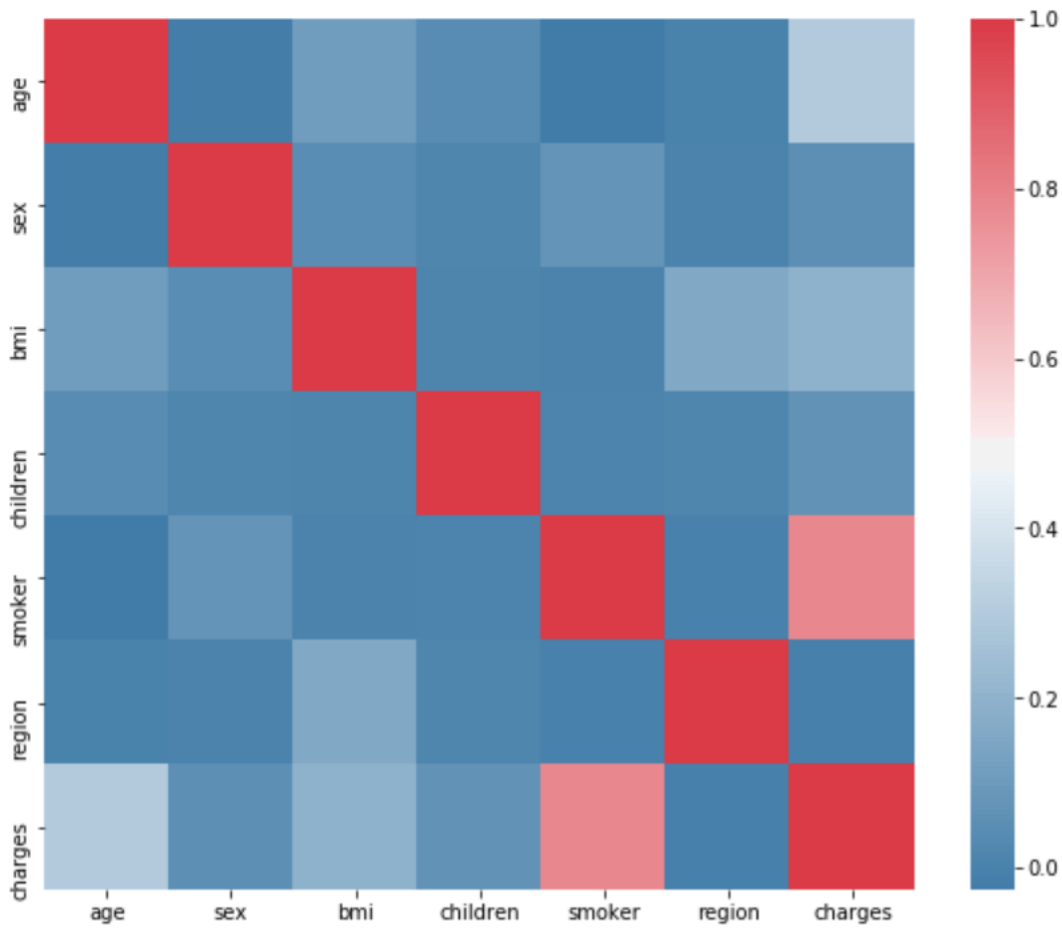
```python
data.corr()['charges'].sort_values()
```

```
region     -0.006208
sex         0.057292
children    0.067998
bmi         0.198341
age         0.299008
smoker      0.787251
charges     1.000000
Name: charges, dtype: float64
```

- A strong correlation is observed only with the fact of smoking the patient; hence, we will deep-dive into smoking in detail.

```
f, ax = pl.subplots(figsize=(10, 8))
corr = data.corr()
sns.heatmap(corr, mask=np.zeros_like(corr, dtype=np.bool), cmap=
sns.diverging_palette(240,10,as_cmap=True),
            square=True, ax=ax)
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x7ffb44a207f0>
```

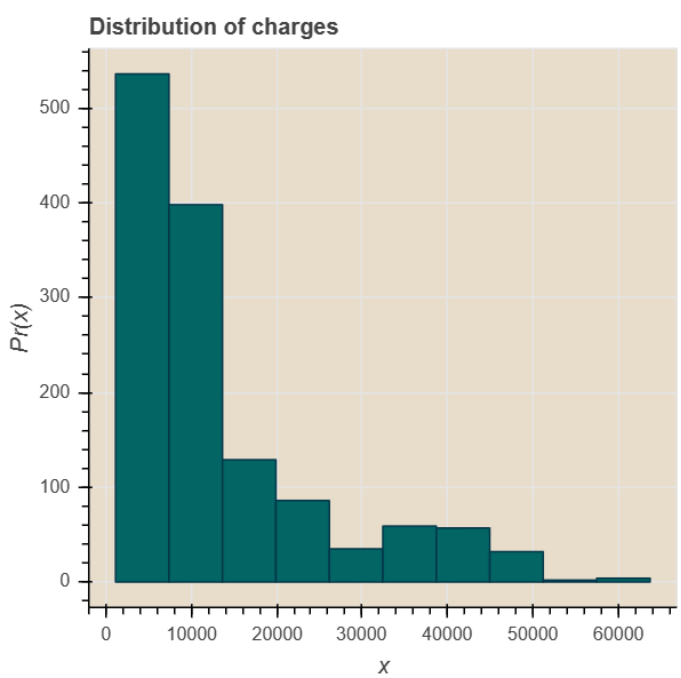


Distribution of Charges:

- This will help to know how much patients spend on treatment on average.
- Although it is not necessary, it will give an overview of the patient's financial condition or medical charges.

```
from bokeh.io import output_notebook, show
from bokeh.plotting import figure
output_notebook()
import scipy.special
from bokeh.layouts import gridplot
from bokeh.plotting import figure, show, output_file
p = figure(title="Distribution of charges",tools="save",
            background_fill_color="#E8DDCB")
hist, edges = np.histogram(data.charges)
p.quad(top=hist, bottom=0, left=edges[:-1], right=edges[1:],
        fill_color="#036564", line_color="#033649")
p.xaxis.axis_label = 'x'
p.yaxis.axis_label = 'Pr(x)'
show(gridplot(p,ncols = 2, plot_width=400, plot_height=400, toolbar_location=None))
```

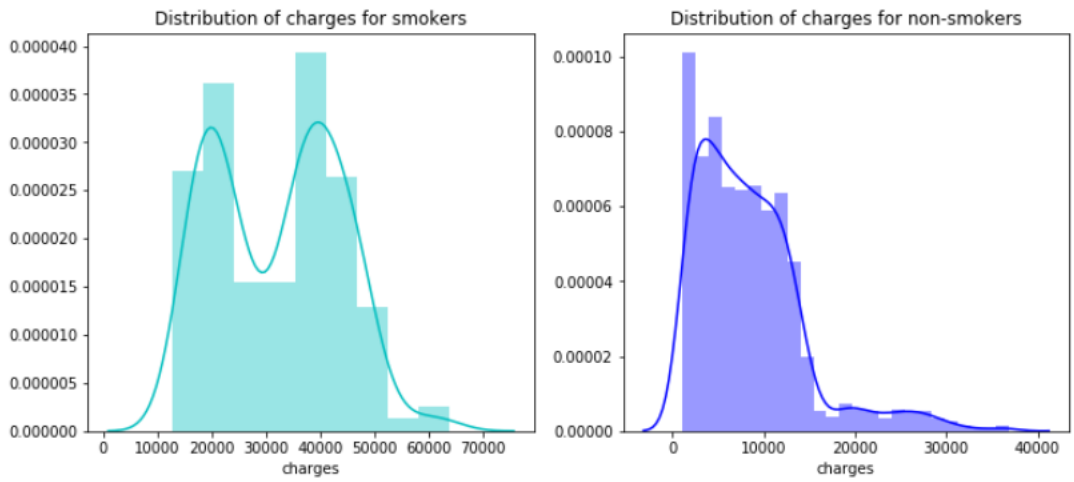
BokehJS 0.13.0 successfully loaded.



```
f= pl.figure(figsize=(12,5))

ax=f.add_subplot(121)
sns.distplot(data[(data.smoker == 1)]["charges"],color='c',ax=a
x)
ax.set_title('Distribution of charges for smokers')

ax=f.add_subplot(122)
sns.distplot(data[(data.smoker == 0)]['charges'],color='b',ax=a
x)
ax.set_title('Distribution of charges for non-smokers')
```

:

```
Text(0.5,1,'Distribution of charges for non-smokers')
```



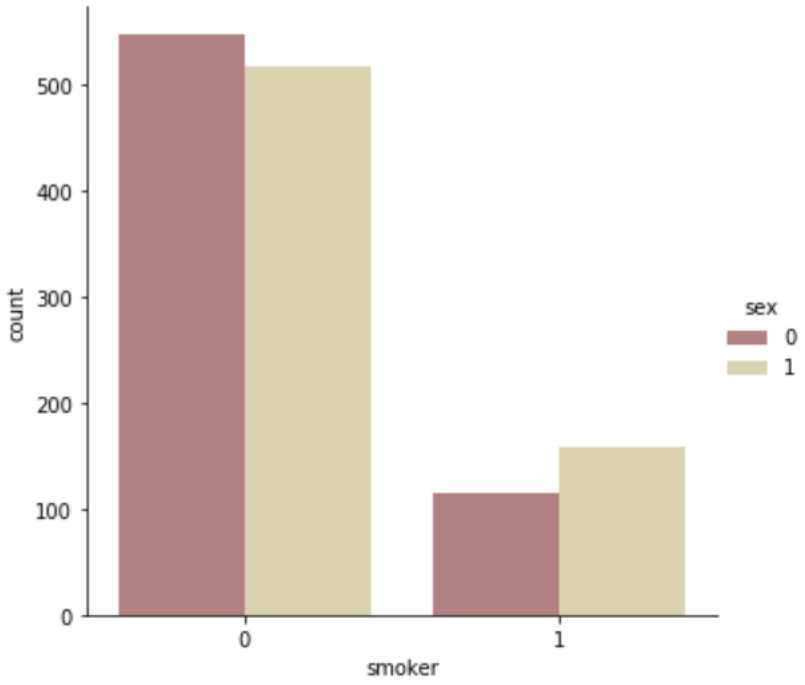
- As we can see, smoking patients spend more on treatment, but not as greater as of non-smoking patients.

Smoking Patients:

Gender-Based Distribution:

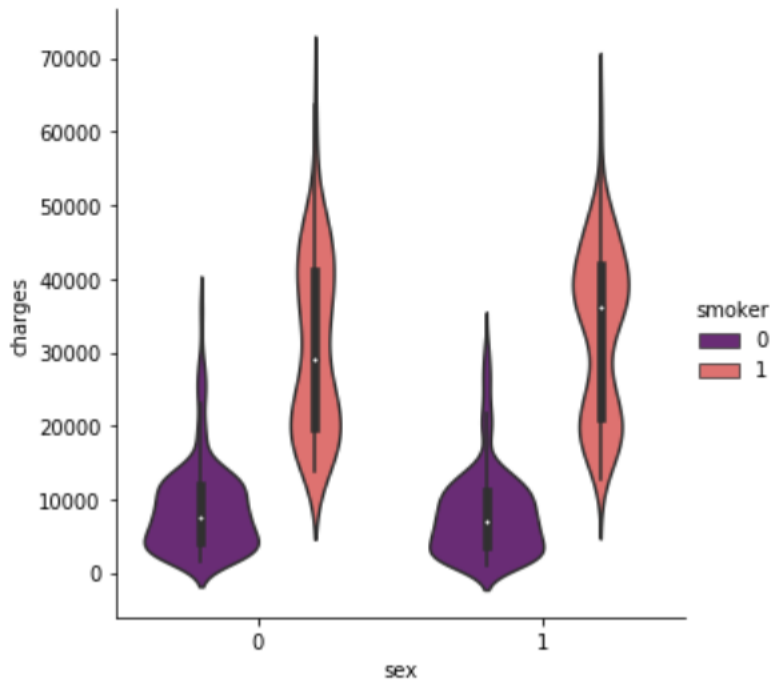
```
sns.catplot(x="smoker", kind="count",hue = 'sex', palette="pin
k", data=data)
```

```
<seaborn.axisgrid.FacetGrid at 0x7ffb42184668>
```



```
sns.catplot(x="sex", y="charges", hue="smoker",
            kind="violin", data=data, palette = 'magma')
```

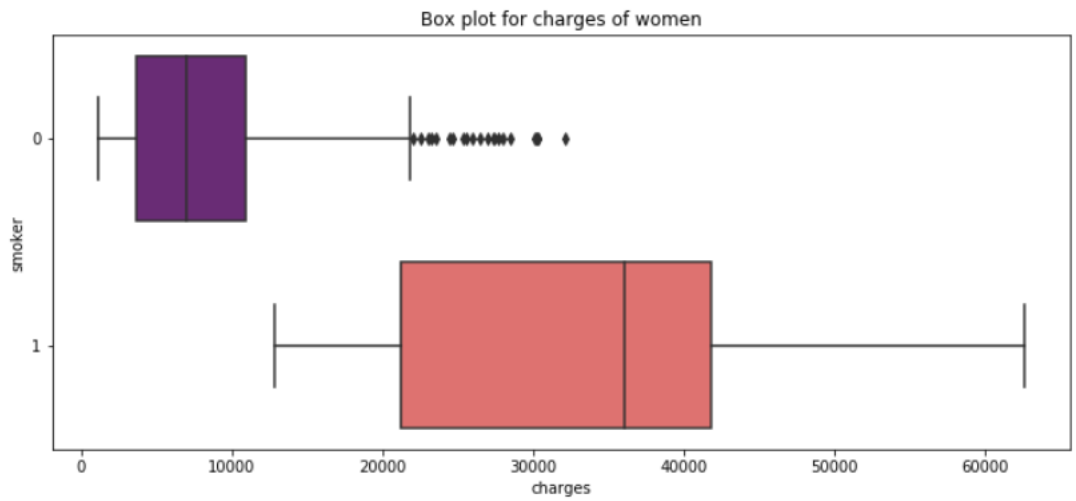
```
<seaborn.axisgrid.FacetGrid at 0x7ffb441e2f28>
```



- If we consider males as “0” and females as “1”, with the help of cat-plots, we can conclude that more men smokers exists than women.
- We may assume that the total cost of treatment in men will be more than in women, given impact of smoking.

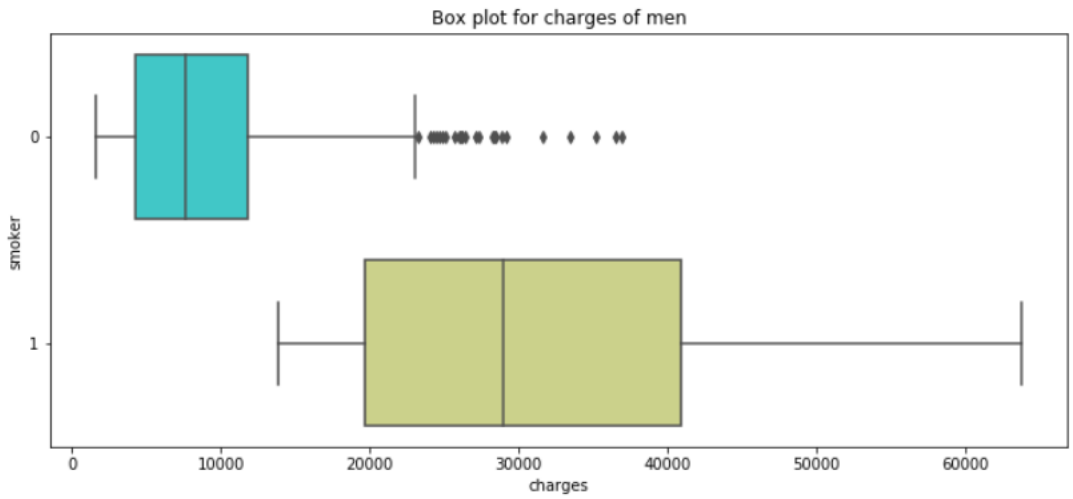
```
pl.figure(figsize=(12,5))
pl.title("Box plot for charges of women")
sns.boxplot(y="smoker", x="charges", data =  data[(data.sex ==
1)] , orient="h", palette = 'magma')
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x7ffb3df2b780>
```



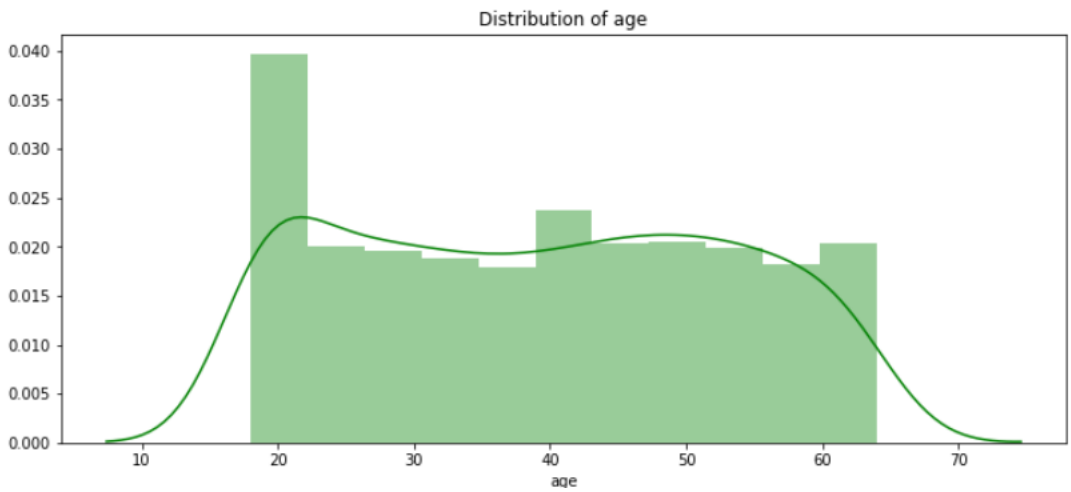
```
pl.figure(figsize=(12,5))
pl.title("Box plot for charges of men")
sns.boxplot(y="smoker", x="charges", data =  data[(data.sex ==
0)] , orient="h", palette = 'rainbow')
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x7ffb3de640f0>
```



Distribution of Age:

```
pl.figure(figsize=(12,5))
pl.title("Distribution of age")
ax = sns.distplot(data["age"], color = 'g')
```

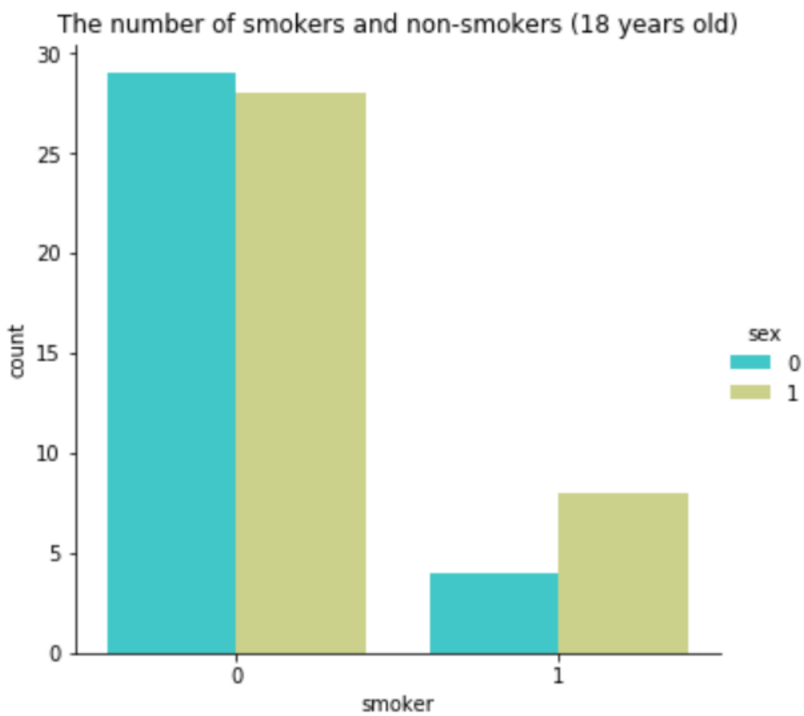


- We are having a few patients under 20 as well while the minimum age of patients should be 18.
- The maximum age is 64.

Determining Smokers at the age of 18:

```
sns.catplot(x="smoker", kind="count",hue = 'sex', palette="rainb
ow", data=data[(data.age == 18)])
pl.title("The number of smokers and non-smokers (18 years old)")
```

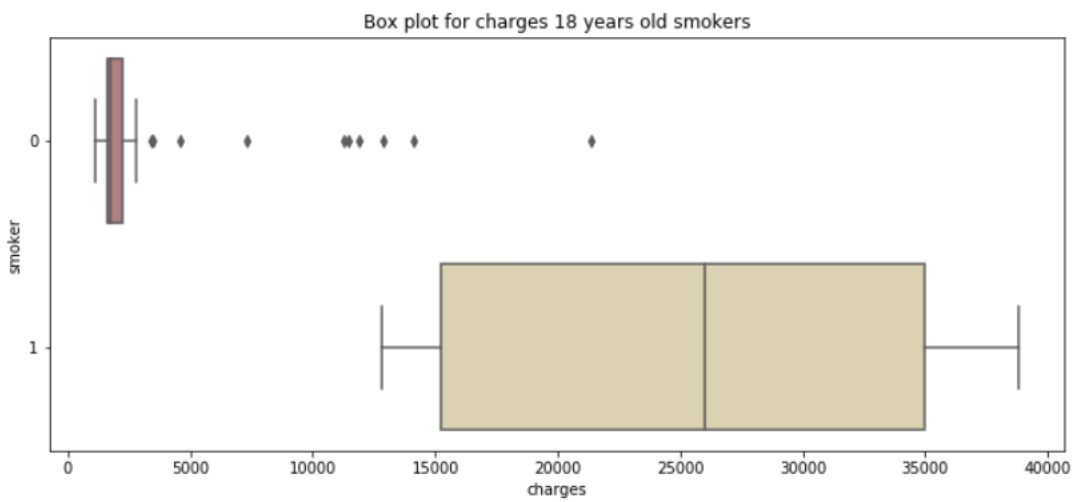
```
Text(0.5,1,'The number of smokers and non-smokers (18 years
old)')
```



Does smoking affect the cost of treatment at the age of 18?

```
pl.figure(figsize=(12,5))
pl.title("Box plot for charges 18 years old smokers")
sns.boxplot(y="smoker", x="charges", data = data[(data.age == 1
8)] , orient="h", palette = 'pink')
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x7ffb3dd2e470>
```

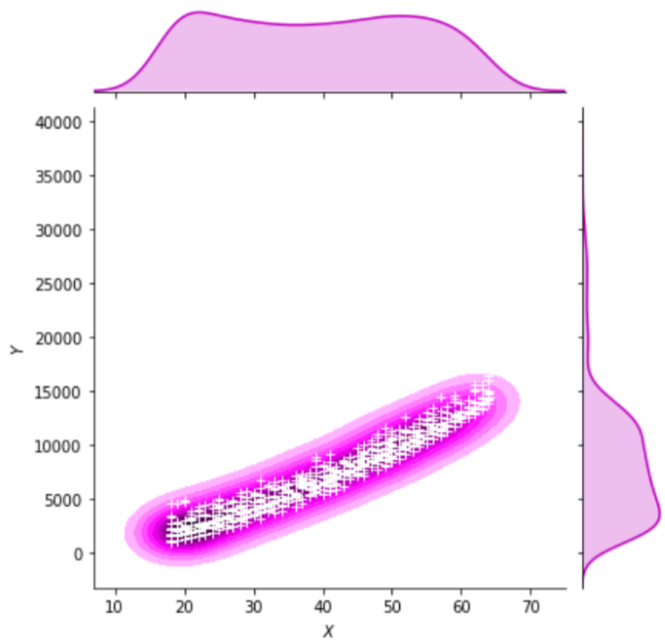


- Even at this young age, smokers spend much more on treatment than non-smokers.
- We may assume that the non-smokers are having some tails is because of serious diseases or accidents.

How treatment cost depends on age of smokers and non-smokers patients?

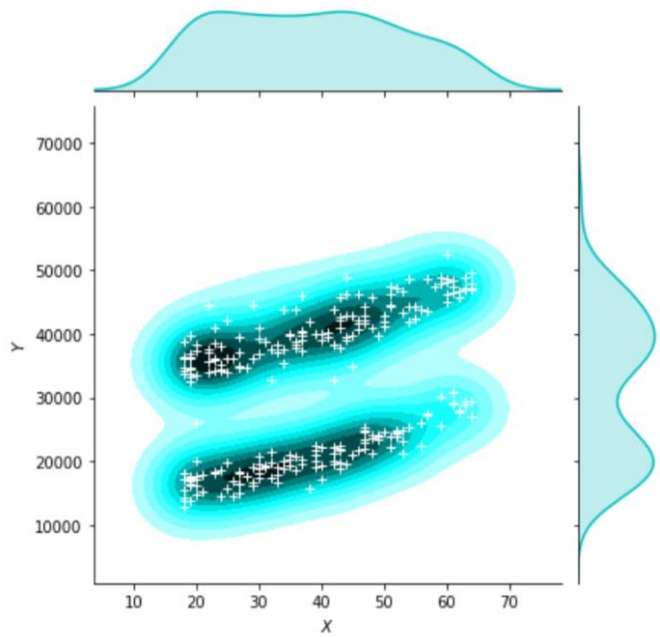
```
g = sns.jointplot(x="age", y="charges", data = data[(data.smoker == 0)],kind="kde", color="m")
g.plot_joint(pl.scatter, c="w", s=30, linewidth=1, marker="+")
g.ax_joint.collections[0].set_alpha(0)
g.set_axis_labels("$X$", "$Y$")
ax.set_title('Distribution of charges and age for non-smokers')
```

```
Text(0.5,1,'Distribution of charges and age for non-smokers')
```

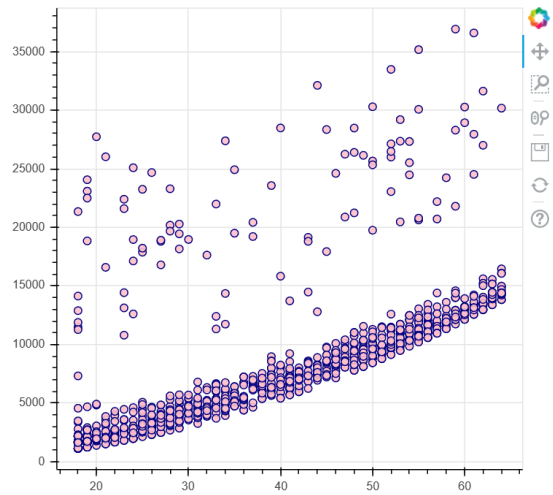


```
g = sns.jointplot(x="age", y="charges", data = data[(data.smoker == 1)],kind="kde", color="c")
g.plot_joint(pl.scatter, c="w", s=30, linewidth=1, marker="+")
g.ax_joint.collections[0].set_alpha(0)
g.set_axis_labels("$X$", "$Y$")
ax.set_title('Distribution of charges and age for smokers')
```

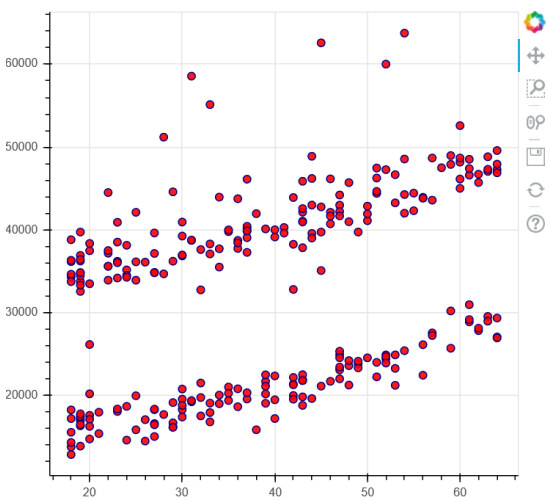
```
Text(0.5,1,'Distribution of charges and age for smokers')
```



```
#non - smokers
p = figure(plot_width=500, plot_height=450)
p.circle(x=data[(data.smoker == 0)].age,y=data[(data.smoker == 0)].charges, size=7, line_color="navy", fill_color="pink", fill_alpha
=0.9)

show(p)
```

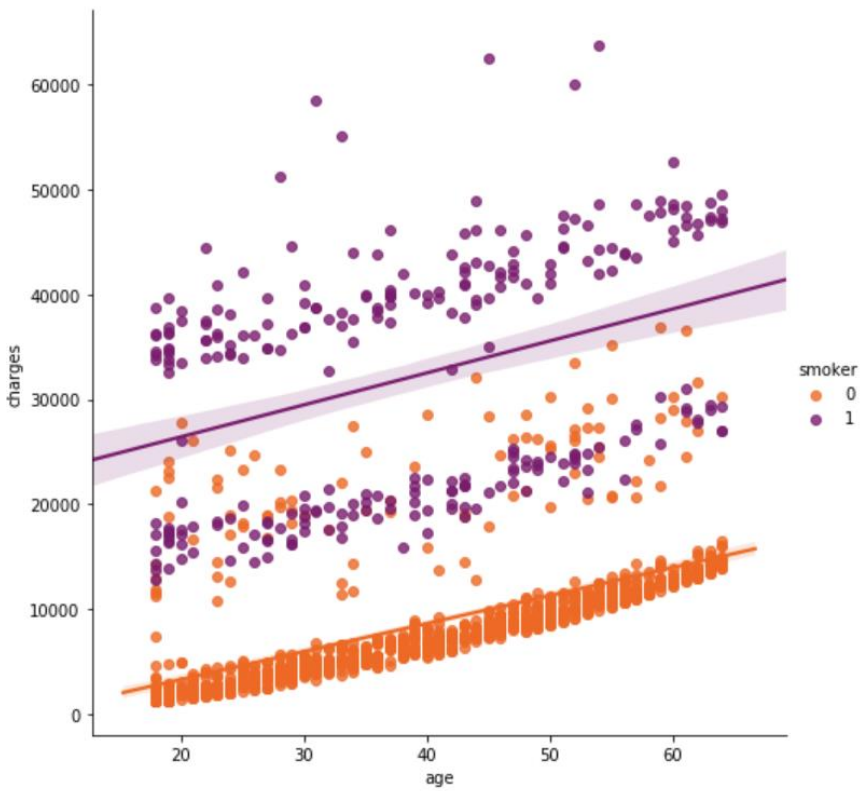



```
#smokers
p = figure(plot_width=500, plot_height=450)
p.circle(x=data[(data.smoker == 1)].age,y=data[(data.smoker == 1)].charges, size=7, line_color="navy", fill_color="red", fill_alpha=
0.9)
show(p)
```



```
sns.lmplot(x="age", y="charges", hue="smoker", data=data, palette = 'inferno_r', size = 7)
ax.set_title('Smokers and non-smokers')
```

```
Text(0.5,1,'Smokers and non-smokers')
```

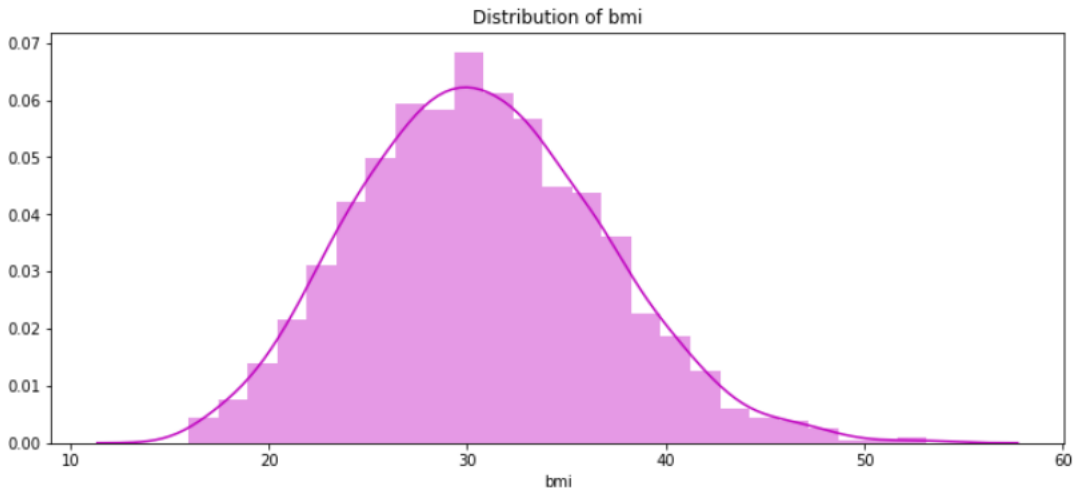


- The cost of treatment increases with age for the non-smokers, which makes sense.
- Smokers do not have such dependence.
- Such a strong effect of smoking on the cost of treatment would be more logical to judge having a set of data with many records and signs.

Distribution of Diet/BMI?

- Let us pay attention to BMI (Body Mass Index) and let us see how the diet affects the cost of treatment in patients.

```
pl.figure(figsize=(12,5))
pl.title("Distribution of bmi")
ax = sns.distplot(data["bmi"], color = 'm')
```

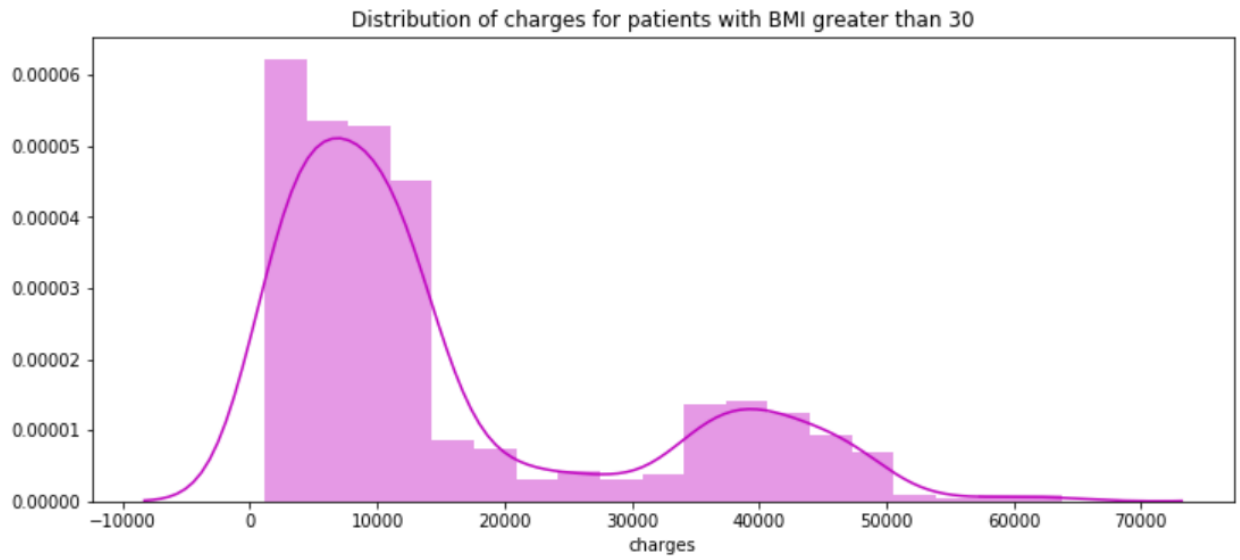


- Average BMI in patients is 30.
- With a value equal to 30 starts obesity.

Distribution of charges for patients with BMI:

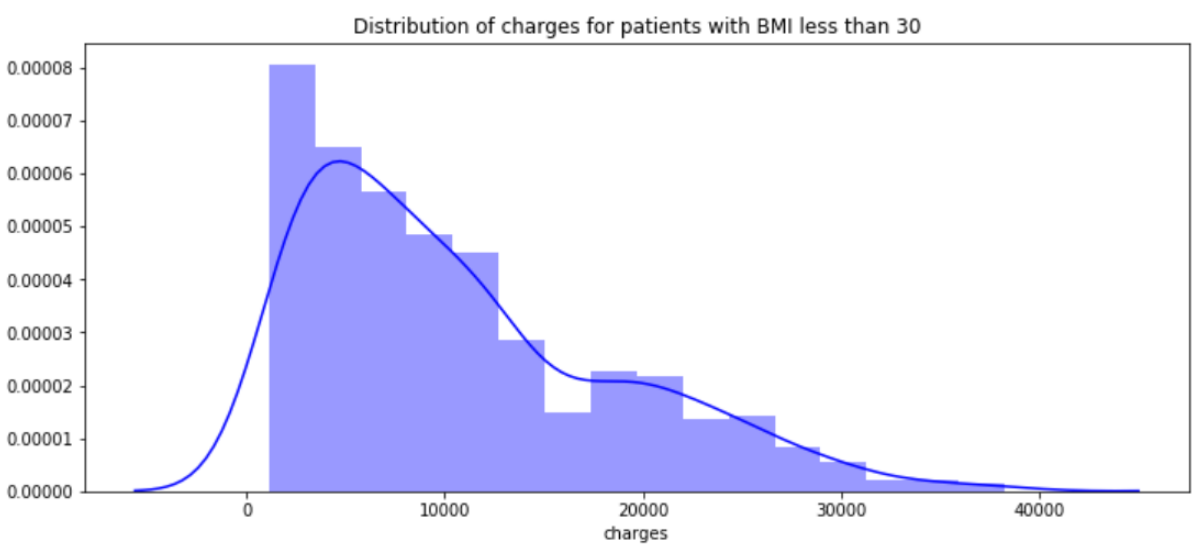
Greater than 30:

```
pl.figure(figsize=(12,5))
pl.title("Distribution of charges for patients with BMI greater than 30")
ax = sns.distplot(data[(data.bmi >= 30)]['charges'], color = 'm')
```



Less than 30:

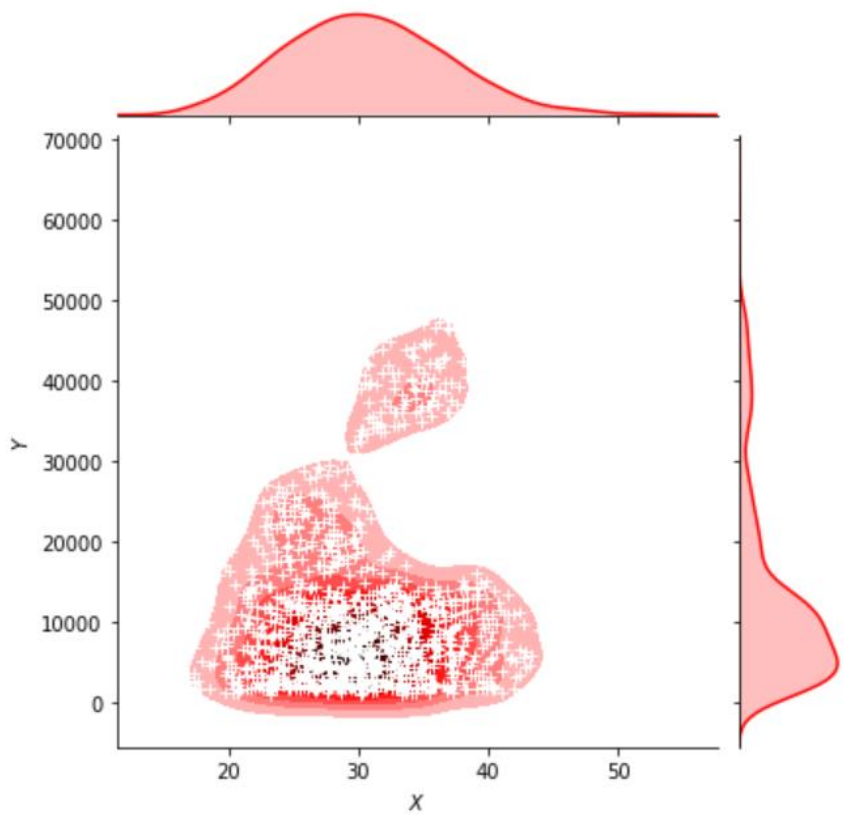
```
pl.figure(figsize=(12,5))
pl.title("Distribution of charges for patients with BMI less than 30")
ax = sns.distplot(data[(data.bmi < 30)]['charges'], color = 'b')
```



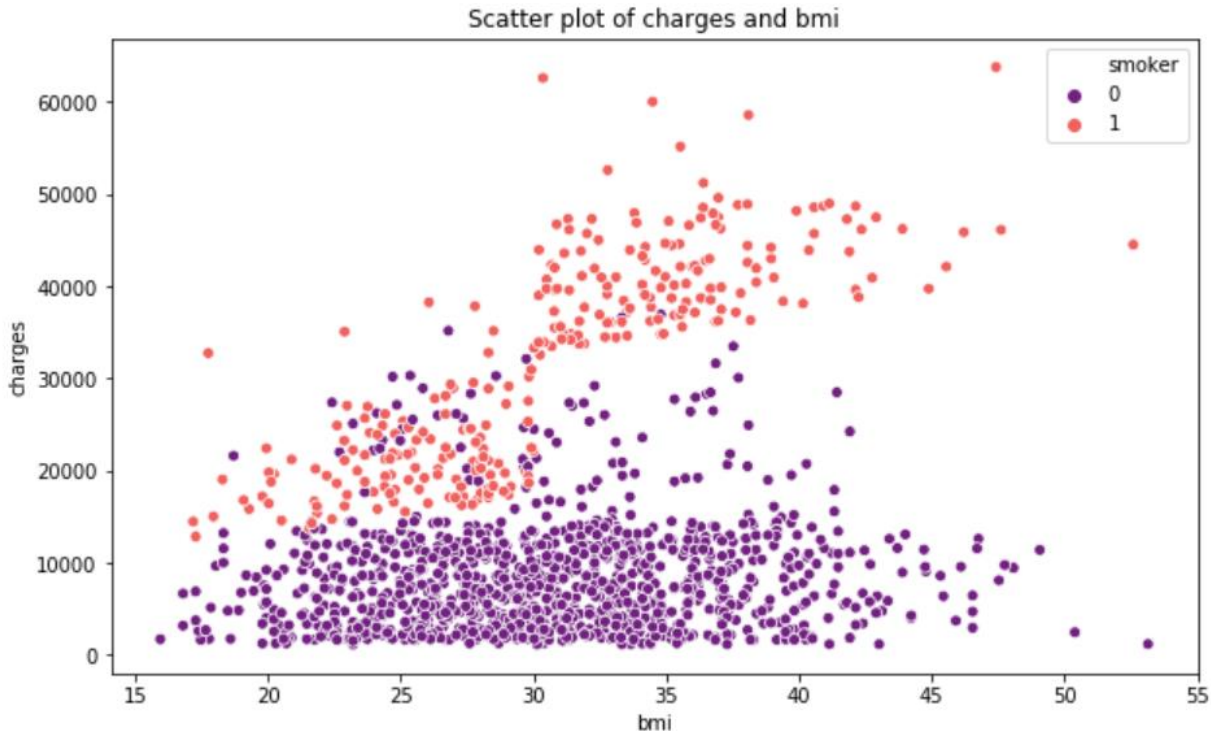
- Hence, patients with BMI above 30 spend more on treatment.

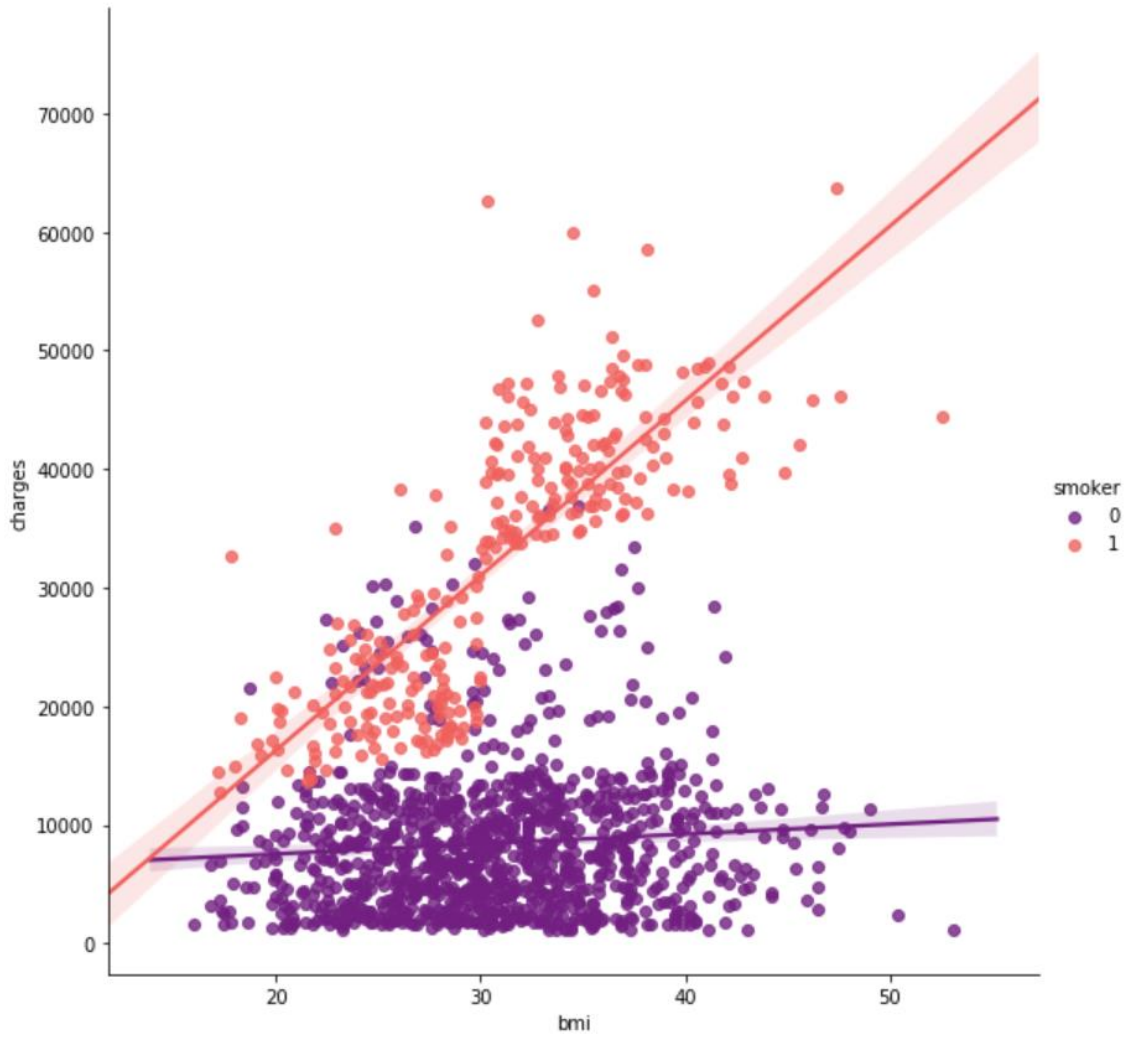
```
g = sns.jointplot(x="bmi", y="charges", data = data,kind="kde", color="r")
g.plot_joint(pl.scatter, c="w", s=30, linewidth=1, marker="+")
g.ax_joint.collections[0].set_alpha(0)
g.set_axis_labels("$X$", "$Y$")
ax.set_title('Distribution of bmi and charges')
```

```
Text(0.5,1,'Distribution of bmi and charges')
```




```
pl.figure(figsize=(10,6))
ax = sns.scatterplot(x='bmi',y='charges',data=data,palette='magma',hue='smoker')
ax.set_title('Scatter plot of charges and bmi')

sns.lmplot(x="bmi", y="charges", hue="smoker", data=data, palette = 'magma', size = 8)
```


Scatter plot of charges and bmi
smoker
0
1
charges
bmi
0
10000
20000
30000
40000
50000
60000
15
20
25
30
35
40
45
50
55

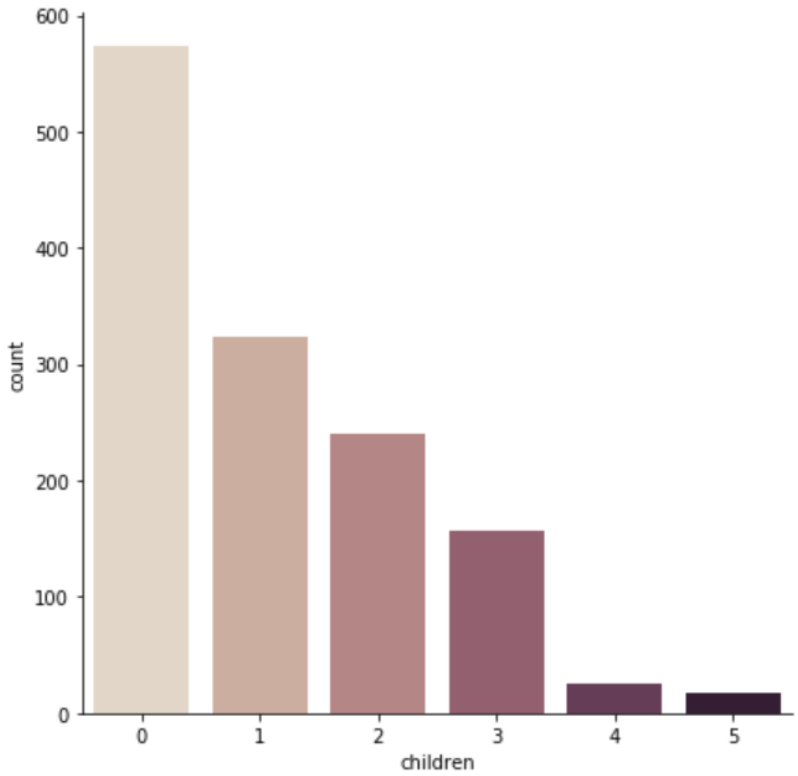


Invigilate Children:

- Figuring out how many children our data set contains.

```
sns.catplot(x="children", kind="count", palette="ch:.25", data=data, size = 6)
```

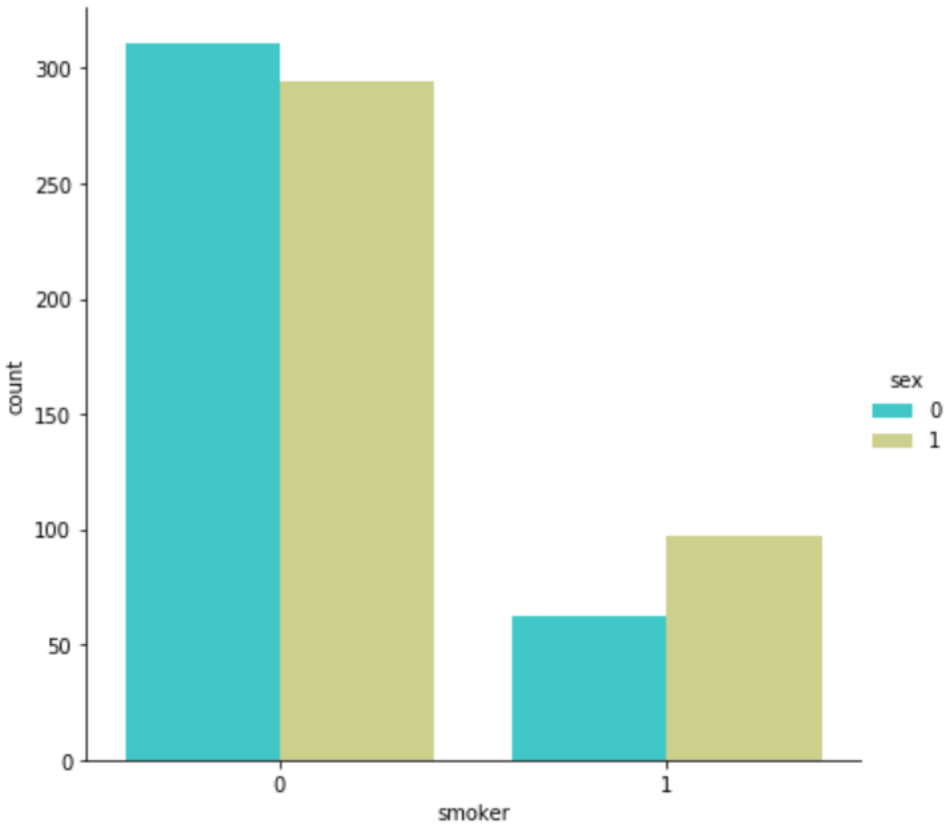
```
<seaborn.axisgrid.FacetGrid at 0x7ffb3d878668>
```



- Most patients do not have children.
- Wondering if people who have children smoke.

```
sns.catplot(x="smoker", kind="count", palette="rainbow",hue = "sex",
            data=data[(data.children > 0)], size = 6)
ax.set_title('Smokers and non-smokers who have childrens')
```

```
Text(0.5,1,'Smokers and non-smokers who have childrens')
```



- Yet, the non-smoking parents are much more.

Prediction using Regression:

Libraries (as mentioned in the beginning):

```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import PolynomialFeatures
from sklearn.metrics import r2_score,mean_squared_error
from sklearn.ensemble import RandomForestRegressor
```

Linear Regression:

```
x = data.drop(['charges'], axis = 1)
y = data.charges

x_train,x_test,y_train,y_test = train_test_split(x,y, random_state = 0)
lr = LinearRegression().fit(x_train,y_train)

y_train_pred = lr.predict(x_train)
y_test_pred = lr.predict(x_test)

print(lr.score(x_test,y_test))
```

```
0.7962732059725786
```

- 79.62% accuracy seems good, especially when doing without data normalization.

Adding Polynomial Features:

```
X = data.drop(['charges','region'], axis = 1)
Y = data.charges



quad = PolynomialFeatures (degree = 2)
x_quad = quad.fit_transform(X)

X_train,X_test,Y_train,Y_test = train_test_split(x_quad,Y, random_state = 0)

plr = LinearRegression().fit(X_train,Y_train)

Y_train_pred = plr.predict(X_train)
Y_test_pred = plr.predict(X_test)

print(plr.score(X_test,Y_test))
```

```
0.8849197344147235
```

- 88.49% accuracy is where our model predicts the cost of treatment of patients on the test-dataset given training dataset.
- We could limit ourselves to creating two or three polynomial features for better results, but this dataset might not suit that case.

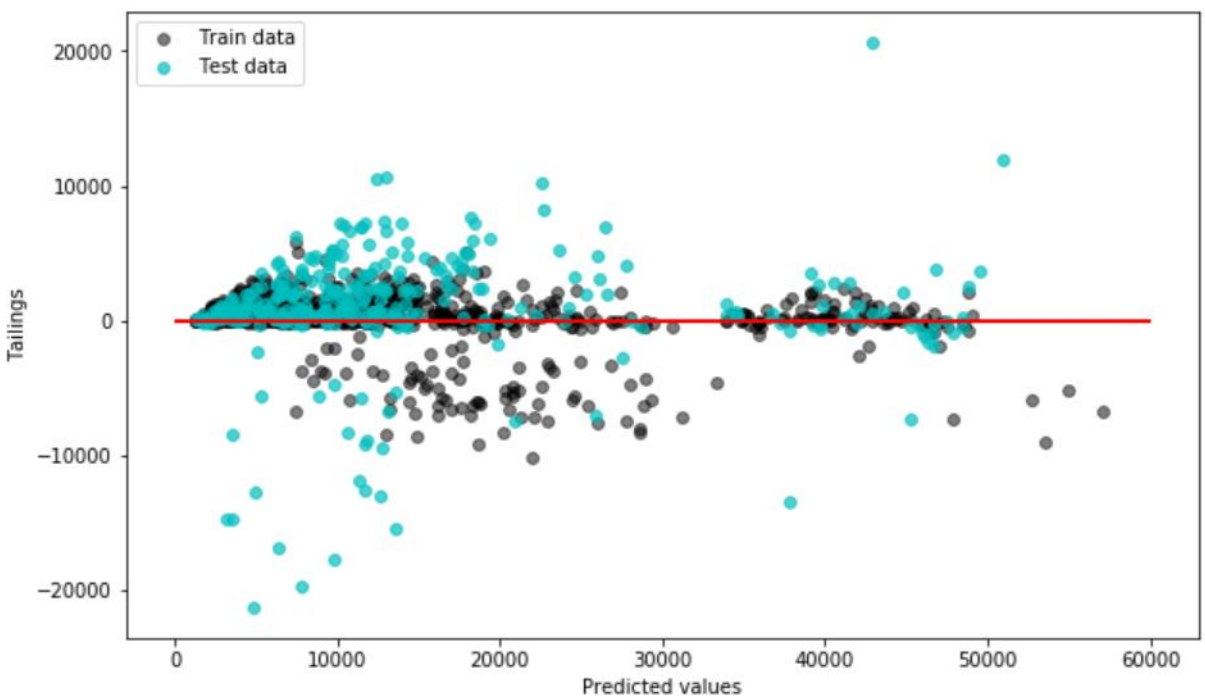
Random-Forest Regression:

```
forest = RandomForestRegressor(n_estimators = 100,
                               criterion = 'mse',
                               random_state = 1,
                               n_jobs = -1)
forest.fit(x_train,y_train)
forest_train_pred = forest.predict(x_train)
forest_test_pred = forest.predict(x_test)

print('MSE train data: %.3f, MSE test data: %.3f' % (
mean_squared_error(y_train,forest_train_pred),
mean_squared_error(y_test,forest_test_pred)))
print('R2 train data: %.3f, R2 test data: %.3f' % (
r2_score(y_train,forest_train_pred),
r2_score(y_test,forest_test_pred)))
```

```
MSE train data: 3729086.094, MSE test data: 19933823.142
R2 train data: 0.974, R2 test data: 0.873
```

Visualizing the test and train data:

```
pl.figure(figsize=(10,6))

pl.scatter(forest_train_pred,forest_train_pred - y_train,
          c = 'black', marker = 'o', s = 35, alpha = 0.5,
          label = 'Train data')
pl.scatter(forest_test_pred,forest_test_pred - y_test,
          c = 'c', marker = 'o', s = 35, alpha = 0.7,
          label = 'Test data')
pl.xlabel('Predicted values')
pl.ylabel('Tailings')
pl.legend(loc = 'upper left')
pl.hlines(y = 0, xmin = 0, xmax = 60000, lw = 2, color = 'red')
pl.show()
```



Conclusion:

- Smoking patients, especially men, are paying more than non-smoking and women patients.
- There are a few patients who smoke at the age of 18 and do pay a lot of money for their treatments.
- As the age increases, the non-smoking patients' treatment cost increases.
- Most patients' BMI comes close to 30, yet above 30 spends more on treatment.
- Hence, a good health depends on a healthy body and mind.